QuantPilot BTC: Complete Project Guide

Version 2026-10-21 · BTCUSDT perpetual futures research and paper trading · Never places real orders

≈50%
direction accuracy: a coin flip, honestly reported

84–87%
live coverage of the 80% price ranges (target 80%)

190+
pre-registered strategy configurations tested

0
paper trades: the model was never confident enough

QuantPilot · BTCUSDT USD-M perpetual · research and paper simulation only · no real-money execution

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1. Executive summary

QuantPilot is a local research platform for Bitcoin perpetual futures (BTCUSDT on Binance USD-M). It runs on one computer, collects market data around the clock, researches what (if anything) predicts Bitcoin, records forecasts and scores them against what actually happened, answers questions in plain language, and runs a simulated (paper) trading account.

What it found, honestly:

Question	Answer
Can it predict whether BTC goes up or down in the next 1 to 24 hours?	**No.** Every model and 186 pre-registered strategies failed to beat a coin flip after costs.
Can it predict how much BTC will move?	**Yes.** 1h and 4h price ranges and next-day volatility beat simple baselines and hit their targets live.
Do support/resistance trade plans make money?	**No.** Every entry rule lost after costs; a coin-flip side did as well.
Has the paper account traded?	**No.** Its model was never confident enough. That is correct behaviour, not a bug.

Current accuracy (2026-10-21):

- Direction: about 50%. The paper model scored 50.7% on its unseen test months, and the live forecasts match a coin flip.
- 80% price ranges: 87% (1h) and 84% (4h) of outcomes landed inside, against an 80% target.

Why this project is valuable anyway: it shows how to do quantitative research properly. It uses real data engineering, leakage-free validation, overfitting controls and pre-registration, and it honestly reports a negative result instead of an impressive but fake backtest.

2. What QuantPilot is and is not

It is:

- a 24/7 data collector;
- a research lab with strict anti-overfitting rules;
- a forecaster that scores itself;
- a question-answering assistant;
- a paper (simulated) trading account.

It is not a money-making bot. There is no code that can send an order to an exchange. Every "trade" is simulated in a local ledger, and research forecasts, strategy tests and chat answers never place or size trades.

3. How it works, end to end

Step	What happens	Where
1. Collect	Every 5 seconds the collector stores new 5-minute candles. History is backfilled to 5 years. Funding, open interest, long/short ratios, mark/index prices, Fear & Greed, macro releases and news are collected on their own schedules. Liquidations and order-book snapshots are recorded live.	`scripts/data_supervisor.py`, `collector.py`, `history.py`, `news.py`, `macro.py`, `forward_data.py`
2. Store	PostgreSQL holds market data. SQLite holds the paper ledger, forecasts, answers and chat.	`market_store.py`, `store.py`
3. Compute features	106 indicators are calculated from completed 5-minute bars, using only information known at that moment.	`features.py`, `context.py`
4. Decide (paper)	Every 15 minutes the frozen XGBoost model predicts the 60-minute direction; the risk gates decide trade or no trade.	`worker.py`, `engine.py`
5. Forecast (research)	Every 15 minutes the shadow forecaster records 1h/4h/24h direction probabilities, 80% ranges, drop risk and volatility. They are scored when their horizon passes.	`forecaster.py`
6. Answer	Questions are parsed into a canonical form and answered with validated methods; Claude explains the answer; everything is scored later.	`questions.py`, `outlook.py`, `answers.py`, `ask_ai.py`, `chat.py`
7. Serve	A FastAPI server feeds the dashboard, the MCP tools for Claude, and the command line.	`api.py`, `services.py`, `mcp_server.py`, `mcp_proxy.py`, `cli.py`
8. Show	A React dashboard with seven pages.	`frontend/src/`

Background service threads (one process):

- collector (every 5 s);
- history backfill (every 30 min);
- paper worker (minutes :02, :17, :32 and :47);
- news (every 5 min);
- macro calendar (hourly);
- shadow forecasts (every 15 min);
- outlook refresh and answer scoring (hourly / every 15 min);
- live liquidations and order book (every minute).

4. Technology stack

Layer	Technology
Language	Python 3.11 (uv for packages), TypeScript
API	FastAPI + uvicorn on 127.0.0.1:8000; write endpoints accept only this computer

Layer	Technology
Databases	PostgreSQL 18 (port 55432) for market data; SQLite for the ledger, forecasts, answers and chat; SQLAlchemy 2, psycopg 3, Alembic migrations
Data	pandas, numpy, pyarrow/Parquet, numba
Machine learning	scikit-learn (logistic regression, calibration, isotonic regression), XGBoost, SHAP explanations, SciPy statistics, custom EWMA and GARCH(1,1)
Networking	httpx (REST), websockets (live liquidations)
AI	Claude Code CLI (Claude Sonnet 5.5) and Codex CLI, both using subscription logins only (never billed API keys); MCP (Model Context Protocol) tools
Frontend	React 19, Vite 6, Tailwind CSS 4, TradingView Lightweight Charts, Recharts, lucide icons
Testing	pytest (334 tests), Playwright browser tests (26 passing, 2 skipped), Node structure checks
Platform	Windows 11 laptop, Node 22, PowerShell and Git Bash, git (branches `main` and `trading-agent`)

5. Data

Data	Source	Coverage
5-minute candles (open, high, low, close, volume, taker buy volume)	Binance public archives + REST API	2021-10 → now, 525,888 of 525,888 bars, no gaps
Mark and index price	Binance	same
Funding rate (every 8 hours)	Binance	same; archives from 2020-01 for research
Open interest; long/short ratios (all accounts, top accounts, top positions)	Binance metrics	same, about 81% of hours
Fear & Greed index	Alternative.me	daily
Macro calendar: CPI, jobs report, FOMC, with first-release values	FRED, ALFRED, Federal Reserve	198 events from 2021
News	Fed, BLS and CoinDesk RSS, analysed by Claude and Codex	from 2026-10-19 (never backfilled, to avoid hindsight)
Spot BTCUSDT daily/hourly, Coinbase BTC-USD, Deribit DVOL (implied volatility)	Binance, Coinbase, Deribit	daily from 2017-08; hourly from 2021-09
Liquidations (live only)	Binance websocket	from 2026-10-20
Order-book depth, 1,000 levels each minute (live only)	Binance REST	from 2026-10-20

6. Indicators and features (106 inputs)

All features use only completed bars and data available at that moment, so nothing peeks into the future.

- **Returns:** 5 minutes, 15 minutes, 1 hour, 4 hours, 24 hours.
- **Trend:**
  - EMA 9/21/50/200 and SMA 20/50/200, as distance and slope;
  - EMA crossover;
  - MACD and MACD histogram;
  - ADX.
- **Momentum:** RSI 7/14/21, Stochastic RSI, Stochastic, Williams %R, CCI, rate of change.
- **Volatility:** ATR, ATR as a fraction of price, realized volatility, Bollinger position and bandwidth, Keltner position, compression, range expansion.
- **Volume:** relative volume, volume change, volume z-score, OBV change, MFI, accumulation/distribution, VWAP distance, price-volume.
- **Candles:** body, upper and lower wick, close location, inside bar, engulfing.
- **Structure:**
  - 20- and 288-bar channel high/low, breakouts and retests;
  - swing high/low distance;
  - previous day and week high/low;
  - pivot, R1 and S1.
- **Multi-timeframe:** RSI, EMA 21/50 distance, ATR fraction, MACD and ADX on 15m, 1h and 4h, plus timeframe alignment.
- **Derivatives:** open interest and its change, price vs OI, the three long/short ratios, funding rate and its change, mark-index premium, taker imbalance, signed taker volume.
- **Sentiment:** Fear & Greed, its change, extreme fear or greed flags and how long they persist.
- **Time and regime:** hour of day (sine and cosine), weekend, regime.

7. Models and algorithms

7.1 The paper-trading model (frozen release `24c8238525d4`)

- **Algorithm:** XGBoost classifier with sigmoid calibration, so that "60%" really means 60%.
- **Question:** will BTC be higher in 60 minutes?
- **Training:**
  - every 5 minutes from 2024-10 to 2026-09;
  - 14 expanding walk-forward folds, with a 60-minute gap between training and test windows to stop overlapping labels leaking (a purge);
  - the final two months held back untouched.
- **Feature selection:** of the 106 inputs, training picked the 40 most useful. All 40 were price and technical indicators; open interest, ratios, funding and sentiment were offered but never chosen.
- **Largest recent influences (SHAP):** 24-hour return, 1-hour RSI, 288-bar channel low, EMA-200 slope, pivot distance, VWAP distance.
- **Result on the untouched holdout:** ROC-AUC 0.539, balanced accuracy 50.65%, zero trades.

7.2 The multi-horizon research harness

- **Direction (1h, 4h, 24h):** logistic regression and XGBoost on five feature groups (price, technical, derivatives, full, full + macro), with expanding or rolling-90-day training.

- **Return ranges:** ridge regression and quantile XGBoost (10th, 50th and 90th percentiles).
- **Large-drop risk (−5% within 24 hours):** XGBoost classifier.
- **Volatility:** EWMA vs GARCH(1,1), scored with QLIKE.
- **Selection:** on validation data only. The test data only confirms or rejects the choice.

7.3 Shadow forecaster

- Refits the winning ("champion") model for each horizon once a day on the latest two years of data.
- Records forecasts every 15 minutes, and scores each one when its horizon passes.
- It never trades.

7.4 Outlook question answering

- **Direction:** a "similar situations" estimator (nearest past market states). It failed validation, so the honest base rate is shown instead.
- **Touch probability** ("will BTC hit $X by T?"): a volatility barrier (reflection) formula, corrected with isotonic calibration. It passed validation but only weakly.
- **80% ranges:** volatility-scaled historical moves, validated at every horizon except 12 hours.
- **Claude's role:** explains the answer and gives its own probability, which is scored separately. Claude never changes the numbers.

7.5 Selective strategy research

- **What was tested:** 10 rule families plus 2 machine-learning filters (meta-labeling), for intraday (hours) and swing (days to weeks) trading:
  - breakouts;
  - momentum agreement;
  - funding crowding;
  - volatility squeeze;
  - Coinbase premium;
  - flush-out rebounds;
  - crowding fades;
  - session momentum;
  - taker flow.
- **How calls are scored:** every call is scored as a real trade: take-profit, stop or time limit, after fees and funding.

7.6 Trading agent: market context, trade plans, simulator

- **Market context:** one snapshot of everything at a moment in time: indicators on 4 timeframes, confirmed support and resistance, previous day and week levels, positioning, funding, the macro calendar and news.
- **Trade plans:**
  - stop just beyond the nearest support/resistance (1.5 to 4 ATR away);
  - target at the next opposing level, within the expected 24-hour range;
  - reward:risk of at least 1.5 after costs;
  - size risking 0.25% of the account.
- **Futures simulator (`futures-v1`):**
  - fills at the next bar's open;
  - if a bar touches both stop and target, the stop counts;

- liquidation checked at the mark price, and funding applied;
- stops can only tighten;
- 24-hour time limit;
- halts after a 2% daily loss or a 10% drawdown.

8. Paper trading rules

When it trades. Every 15 minutes, all of these must be true:

1. The model's probability that BTC rises in 60 minutes is **60% or more (long)** or **40% or less (short)**.
2. The expected move (probability × average gain − (1 − probability) × average loss) is larger than costs (0.16% round trip) + funding + a 0.05% buffer.
3. The data is fresh (under 10 minutes old) and complete.
4. No position is already open, the account hasn't lost 2% today, and it isn't paused.

How a trade is run:

- it risks 0.25% of the account;
- the stop is 1.5 × ATR away;
- the position is at most 25% of the account;
- it closes after 60 minutes or at the stop.

What happens in practice: the model's confidence stays between about 45% and 55%, so every check ends "no trade: weak forecast, insufficient edge".

9. How overfitting was controlled

Control	What it prevents
Purged walk-forward splits	Training on the future, or on overlapping labels
Selection on validation only; constant base rate wins ties	Choosing models that only look good on the test
Pre-registration (`docs/research/preregistration.md`)	Changing the rules after seeing results
Trial registry (every configuration counted)	Hiding how many attempts were made
Deflated Sharpe ratio	Lucky winners among many attempts
PBO (probability of backtest overfitting)	Configurations that fall apart out of sample
Wilson confidence intervals and break-even hit rates	Mistaking noise for skill
One-shot holdout year (2025-10-20 → 2026-10-19, still untouched)	Testing until something works
Leakage canaries: a future-peeking signal must pass (92%), random ones must fail	Hidden leakage in the test machinery
No historical news backfill	Language models "remembering" what happened next

10. Results in full

10.1 Direction research (221,213 five-minute cutoffs)

Horizon	Champion	PBO	Verdict
1h	base rate	0.25	no edge
4h	base rate	0.25	no edge
24h	base rate	0.75	no edge (one fold looked good, then failed: textbook overfitting)

10.2 What did work

Model	Result
1h 80% range (with macro calendar features)	+3.8% better than the unconditional range, 4 of 4 folds, PBO 0.00
4h 80% range	+1.4%, passes at exactly half the folds
Volatility	EWMA best at 1h and 4h; GARCH(1,1) best at 24h
Touch probability	small but validated improvement (Brier 0.09320 vs 0.09351)

10.3 Five-year 5% move event study

- 143 drops and 175 rallies of 5% or more.
- After a 5% drop, the next 24 hours had a median of +0.44%, vs −0.32% for comparable ordinary hours.
- High volatility, crowded longs and high funding made drops 2 to 4 times more likely, but not predictable enough to trade.
- Only 19 of the 143 drops followed a CPI, jobs or FOMC release, about what chance would give.

10.4 Selective strategies (186 configurations, pre-registered)

- **No strategy passed.**
- **Closest: trend + ML filter (swing, 7-day hold):**
  - 295 calls with a 58.3% hit rate, against a 51.6% break-even;
  - most of that is Bitcoin's 2017–2025 uptrend (always-long scored 52.9%);
  - failed the overfitting check (PBO 0.42).
- **Intraday:** every family lost money after costs.

10.5 Trading-agent trade plans (step B)

Entry rule	Trades (4 years)	Hit rate	Break-even	Account
Trend (1h and 4h agree)	66	42.4%	44.6%	+0.4%
Best plan, either side	380	33.9%	38.9%	−12.1%
Pullback in trend	11	27.3%	40.7%	−0.9%
Coin flip (control)	266	35.7%	38.3%	−5.6%

Walk-forward: 192 trades, a 30.2% hit rate against a 36.9% break-even, −0.23% per trade, PBO 1.0. **Fail.**

Why trade management can't fix this: only 13–14% of stopped-out trades were ever ahead by the full stop distance. When the direction can't be predicted, no exit rule can create profit.

10.6 Live accuracy so far

What	Live result
Research direction forecasts (1h/4h/24h)	Brier score identical to the base rate (about 0.25): no edge
80% ranges	87% (1h) and 84% (4h) of outcomes inside: working
Chat up/down answers	63.6% on 44 answers: luck from a rising day; will drift toward 50%

11. Using QuantPilot

Task	How
Start everything	`node scripts/start.cjs` (restart after code changes: `--restart-services`)
Dashboard	http://localhost:5173
API documentation	http://127.0.0.1:8000/docs
Ask a question	Dashboard top bar (press `/`), the Ask page, or `quantpilot ask "Chance BTC touches 90k this week?"`
Use from Claude Code	`claude mcp add quantpilot -- "C:\Projects\BTC Quant\.venv\Scripts\python.exe" -m quantpilot.mcp_proxy`
Run the tests	`.venv/Scripts/python.exe -m pytest -q -p no:cacheprovider`, then `cd frontend && npx playwright test`
Re-run research	`quantpilot research-data`, `quantpilot research-strategies --output DIR`, `quantpilot research-agent --output DIR`

Dashboard pages:

Page	What's on it
Overview	price, chart, outlook (1h/4h/24h toggle), market pulse, next macro event, headlines, research verdict, system status
Chart	86 indicator studies, 7 timeframes, drawing tools, live stream
Ask	Claude chat and answer scoreboard
Research	findings, forecasts, strategy and trading-agent results, original model evidence
Paper	account, current decision and gates, ledger, audit
News	macro calendar and news with AI analysis
System	readiness, collectors, worker runs, data coverage

API:

- Read endpoints: candles, indicators, positioning, sentiment, coverage, predictions, backtests, portfolio, decisions, news, macro, forecasts, strategy-research, answers, chat threads.
- Write endpoints, this computer only: ask, outlook, chat, AI view.

MCP tools (16):

- market snapshot, indicators, positioning, sentiment, coverage;
- prediction, decision explanation, backtest comparison, paper portfolio;
- news, macro calendar, shadow forecasts, strategy research;
- ask outlook, record AI view, answer scoreboard.

12. Project history

When	Who	What
Before 2026-10-20	Codex	Built the platform: data pipeline and audits, 106 features, walk-forward training, the frozen release model, paper engine, chart workspace, first MCP tools, recovery drills
2026-10-20	Claude	News pipeline with two AI reasoners; macro calendar; 5% event study; multi-horizon research harness; shadow forecaster; outlook Q&A and in-app chat
2026-10-20	Claude	Pre-registered selective strategy research (no edge found); live liquidation and order-book collectors
2026-10-20/21	Claude	Trading agent steps A (context, trade plans, simulator, 7 defect fixes) and B (trade plans tested: no edge)
2026-10-21	Claude	Codebase cleanup; faster API; dashboard redesign; this guide

Every piece of work was reviewed by an independent reviewer, and its findings were fixed test-first.

13. Known limitations

- **Memory:** the laptop (16 GB RAM) runs near its limit; long jobs can be killed and live data has gaps when it sleeps. An always-on machine fixes both.
- **Scope:** one market (BTCUSDT), so no cross-market strategies.
- **Paper engine:** bar-based fills and approximate margin.
- **News:** only days of history, so it can't be evaluated yet.
- **Liquidation data:** a lower bound (the exchange reports at most one per second); about 5.5 hours were missed on 2026-10-20.
- **Backup:** no git remote, so code exists only on this computer.
- **Authorship:** most code was written by AI assistants (Codex, Claude), so the owner should learn the core concepts before presenting it.

14. What's next

1. **Live-data tests (from 2027-04-20):** liquidation-exhaustion and order-book-imbalance tests, already pre-registered.
2. **Reliability:** Docker Compose on an always-on machine; a GitHub repository with automated tests (CI).

3. **MLOps:**

- MLflow experiment tracking and model registry;
  - Grafana monitoring and alerts for data freshness, drift and calibration.

1. **Optional:** an experimental live paper account running the trade plans (step D), clearly labelled as having no edge.
2. **If seeking a real edge:** expand to many coins (relative-strength rotation), options data, or market making, rather than single-asset direction prediction.

Important: MLOps and better infrastructure improve reliability and speed, not prediction accuracy. Accuracy is limited by the information in the data.

15. Using this project on a resume

Frame it as rigorous research, not as a profitable bot.

Suggested bullets:

- Built an end-to-end BTC futures research platform (Python, PostgreSQL, FastAPI, React) with 5 years of exchange data, live websocket collectors and automated data-quality audits.
- Tested 190+ pre-registered trading strategies with purged walk-forward validation, deflated Sharpe ratios and PBO; found no direction edge after costs, but validated volatility and range forecasts (80% ranges at 84–87% live coverage).
- Integrated Claude-powered agents via MCP with read-only tools, and automatically scored every AI and model prediction against a base rate.

Be ready to explain:

- walk-forward validation and purging;
- data leakage;
- Brier score and calibration;
- PBO and the deflated Sharpe ratio;
- why transaction costs kill most strategies;
- why "no trade" is the correct outcome here.

16. Glossary

Term	Meaning
Base rate	How often something happened historically; the honest answer when no model beats it
No edge	No method beat the base rate out of sample
Walk-forward	Train on the past, test on the next period, roll forward
Purge	A gap between training and test data so overlapping outcomes can't leak
Holdout	Data kept unseen until one final test
Pre-registration	Writing rules and pass criteria down before testing
Brier score	Penalty for probability forecasts; lower is better; a coin flip scores 0.25

Term	Meaning
Calibration	Making a "60%" forecast come true about 60% of the time
PBO	Probability of backtest overfitting: how often the best-looking setting falls to the bottom half later
Deflated Sharpe	Chance a result is real after counting every attempt
Hit rate / break-even	Share of profitable calls / share needed to make zero
Triple barrier	A trade ends at take-profit, stop or time limit, whichever comes first
Meta-labeling	A rule picks the direction; a second model decides whether to act
80% range	Interval expected to contain the outcome 8 times out of 10
Funding	Payment between longs and shorts every 8 hours on perpetual futures
Open interest	Total open positions
Liquidation	A position forcibly closed by the exchange
MCP	Model Context Protocol: how Claude calls QuantPilot's tools
Shadow forecast	Recorded and scored but never traded

17. Where everything is

What	Where
Live status document	`progress.md`
This guide	`docs/QuantPilot-Project-Guide.md` and `.pdf`
Research write-ups	`docs/forecast-research.md`, `docs/strategy-research.md`, `docs/trading-agent-research.md`, `docs/event-study.md`
Rules registered before testing	`docs/research/preregistration.md`
Designs and plans	`docs/superpowers/specs/`, `docs/superpowers/plans/`, `docs/prediction-plan.md`
Python code	`src/quantpilot/`
Dashboard code	`frontend/src/`
Tests	`tests/`, `frontend/tests/`
Research artifacts	`artifacts/`
Data caches and backups	`data/`